



Apollo Seiko L-CAT NEO-N (Credit: www.apolloseiko.com)



HAKKO FM-203 (Credit: <https://hakkousa.com>)

solder station. Since PCBs have different components, there is a need to use soldering tips of different shapes, sizes, and temperatures depending on the application. Even using a cartridge type tip may not be easy as it needs to be replaced to continue the soldering process. With recent advancements in technology, soldering stations give the convenience of getting two different tasks done with one station.

The dual port soldering station HAKKO FM-203 comes with another handpiece connected to the other port/channel which has temperature settings and tip shape installed.

IoT solder station. The Internet of things (IoT) soldering station represents the latest evolution of the soldering station connected to the company network that enables full traceability for manual soldering. As soon as the soldering station is integrated into a company network, all mobile devices in the company

network can access and communicate with the soldering stations via web browser. The server based communication concept makes it easier to manage individual soldering stations and makes the process documentable.

For instance, the i-CON TRACE is claimed to be the world's first fully networkable soldering station with integrated wireless local area network (WLAN), Bluetooth, and network card.

The way forward: automation

Soldering becomes more precise and error-free when automated, and is ideal for soldering complex components onto PCBs for a variety of applications in the fields, including automotive, aviation, and telecommunications. The equipment comes as hi-tech integrated packages, while robots come with various features with sturdy construction design to give optimum performance. Both have similar features like high-resolution cameras to support soldering with real-time view, vision alignment, great temperature control, superior thermal feedback system, nitrogen systems, and many more for most challenging solders. Therefore, industry experts believe both robots and soldering equipment are ideal for the soldering process.

“Choosing a soldering equipment or robot is an application-driven query. There are different needs for PCBs coming to the manufacturers. For instance, if a board has more thermal

demands, we look for machines with more pre-heating technologies, and if something has more small components, we look for the availability of micronozzle features. It is also important to consider the size of the mother boards, and selective soldering is a great solution to solder larger boards. However, manual soldering may not be advisable for miniatures used in IoT devices or auto-ancillary equipment,” says Arvind Gopal Joshi, director at Accurex Solutions, a sales and support provider of capital equipment used for pcba.

Joshi suggests looking at specific standards while planning automation. “The automated equipment launched in recent times are designed to meet standards like the IPC 2591 connected factory exchange (CFX),” he says.

According to the IPC global trade association, the IPC 2591 is the open industry standard that supports IPC-CFX. The standard applies to communication between all executable processes in the manufacture of printed board assemblies—automated, semiautomated, and manual—and is applicable to related mechanical assembly and transactional processes.

Despite the high initial expense in setting up both equipment or robots, over time automated soldering proves cost-effective compared to soldering by hand. By using automated machines, both the cost of labour and costs associated with the training and re-training of soldering experts is reduced. Additionally, the increased precision afforded by these machines can ensure better quality work that eliminates the need for rework. Investing in good quality parts and accessories further reduces costs over time.

The automated systems will be more prevalent in the future while they would still be working along with humans, assisting the experts to do their job more efficiently and effectively. **EFY**



i-CON TRACE (Credit: www.kurtzrsa.com)

The author, Abhinaya Kuzhanthaivel, is a technology enthusiast